

**PRAKTIKUM SISTEM OPERASI**

**MODUL 5**

**(EKSPLORASI PROSES XINU)**



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**2026**

## JURNAL 2

1. Jawablah pertanyaan berikut ini:

a. Berapa banyaknya maksimum proses yang ada pada Xinu?

```
#ifndef NPROC
#define NPROC      8
#endif
```

Pada process.h, jumlah maksimum process di atur oleh konstanta NPROC, pada screenshot, ditampilkan bahwa jumlah maksimum proses yang ada pada Xinu sekarang adalah 8.

b. Berapa maksimal panjang nama suatu proses pada Xinu?

```
#define PNMLEN      16      /* Length of process "name" */
```

Pada process.h, panjang nama maksimum sebuah process di atur oleh konstanta PNMLEN dengan nilai default 16.

c. Berapa nilai prioritas awal pada saat proses dibuat?

```
#define INITPRIO    20      /* Initial process priority
```

Pada process.h, nilai prioritas awal suatu process di atur oleh konstanta INITPRIO dengan nilai default 20.

d. Ada berapa total state pada Xinu? Sebutkan!

```
/* Process state constants */
```

```
#define PR_FREE      0      /* Process table entry is unused */
#define PR_CURR      1      /* Process is currently running */
#define PR_READY     2      /* Process is on ready queue */
#define PR_RECV      3      /* Process waiting for message */
#define PR_SLEEP     4      /* Process is sleeping */
#define PR_SUSP      5      /* Process is suspended */
#define PR_WAIT      6      /* Process is on semaphore queue */
#define PR_RECTIM    7      /* Process is receiving with timeout */
```

Ada 8 state, silahkan mengacu ke screenshot diatas.

2. Perintah ps adalah perintah untuk menampilkan statistik process yang berjalan. Source code dari ps tersimpan pada file xsh\_ps.c. Carilah file tersebut dan beri komentar pada 20 baris terakhir di source code tersebut!

```

37     }
38
39     /* Print header for items from the process table */
40
41     //printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
42             // "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
43             // "Stack Ptr", "Stack Size");
44
45     //printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
46             // "----", "-----", "-----", "----", "-----",
47             // "-----", "-----", "-----");
48
49     /* Output information for each process */
50
51     //for (i = 0; i < NPROC; i++) {
52         //prptr = &proctab[i];
53         //if (prptr->prstate == PR_FREE) { /* Skip unused slots */
54             //continue;
55         //}
56         //printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
57                 //i, prptr->prname, pstate[(int)prptr->prstate],
58                 //prptr->prprio, prptr->prparent, prptr->prstkbase,
59                 //prptr->prstkptr, prptr->prstklen);
60     //}
61     //return 0;
62 }

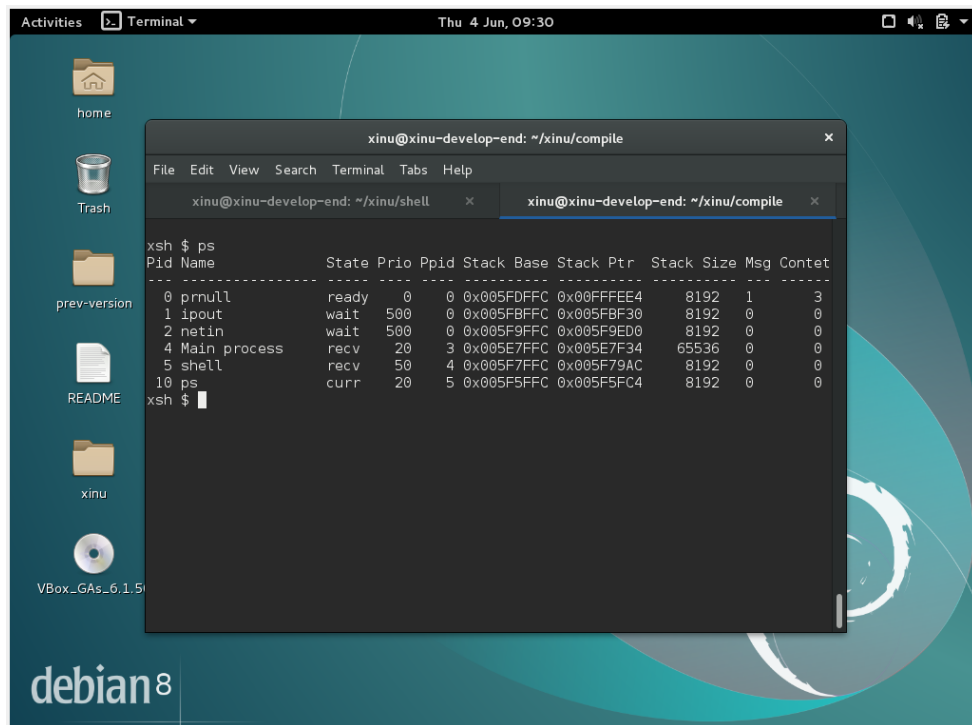
```

- Ubahlah perintah ps (source code: xsh\_ps.c) pada Xinu sehingga menampilkan informasi tambahan berupa kolom yang berisi total message

```

61     //return 0;
62
63     printf("%3s %-16s %5s %4s %4s %10s %-10s %10s %3s %s\n",
64            "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
65            "Stack Ptr", "Stack Size", "Msg", "Content");
66
67     printf("%3s %-16s %5s %4s %4s %10s %-10s %10s %3s %s\n",
68            "----", "-----", "-----", "----", "-----", "-----",
69            "-----", "-----", "-----", "-----");
70
71     /* Output information for each process */
72
73     for (i = 0; i < NPROC; i++) {
74         prptr = &proctab[i];
75         if (prptr->prstate == PR_FREE) { /* Skip unused slots */
76             continue;
77         }
78         printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d %3d %7u\n",
79                i, prptr->prname, pstate[(int)prptr->prstate],
80                prptr->prprio, prptr->prparent, prptr->prstkbase,
81                prptr->prstkptr, prptr->prstklen,
82                prptr->prhasmsg ? 1 : 0,
83                prptr->prhasmsg ? prptr->prmsg : 0);
84     }
85     return 0;
86 }

```



- Ubahlah perintah uptime pada Xinu sehingga menampilkan lamanya Xinu sejak booting hanya dalam satuan menit.

```

Open  *xsh_uptime.c  Save  x
~/xinu/shell

if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
    printf("Use: %s\n\n", args[0]);
    printf("Description:\n");
    printf("\tDisplays time since the system booted\n");
    printf("Options:\n");
    printf("\t--help\t display this help and exit\n");
    return 0;
}

/* Check for valid number of arguments */

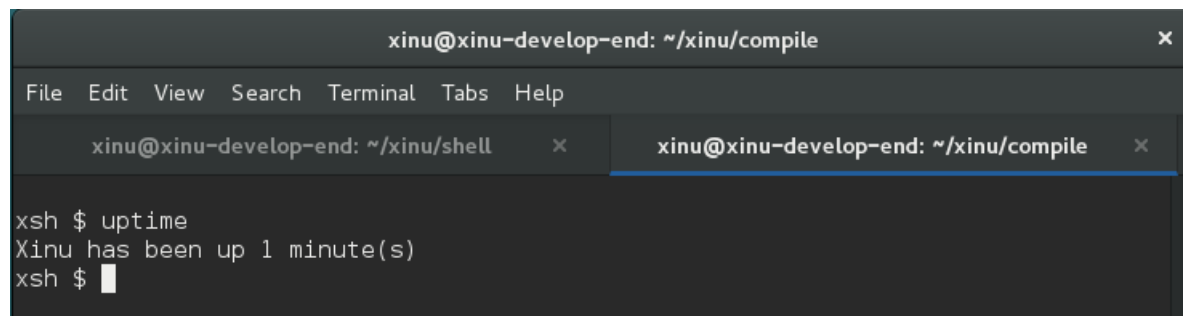
if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
    return 1;
}

mins = clktime / secpermin

printf("Xinu has been up %d minute(s)\n", mins);
//cname(1,20); // tambahkan

return 0;
}
C  Tab Width: 8  Ln 39, Col 9  INS

```



The image shows a terminal window with a dark background. At the top, there is a title bar that reads "xinu@xinu-develop-end: ~/xinu/compile" with a close button on the right. Below the title bar is a menu bar with the options "File", "Edit", "View", "Search", "Terminal", "Tabs", and "Help". Under the menu bar, there are two tabs: "xinu@xinu-develop-end: ~/xinu/shell" and "xinu@xinu-develop-end: ~/xinu/compile", with the latter being the active tab. The main area of the terminal displays the following text: "xsh \$ uptime", "Xinu has been up 1 minute(s)", and "xsh \$ " followed by a cursor.

```
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Tabs Help
xinu@xinu-develop-end: ~/xinu/shell xinu@xinu-develop-end: ~/xinu/compile
xsh $ uptime
Xinu has been up 1 minute(s)
xsh $
```